

Ishaan Mittal

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SUMMARY

Software Engineer focused on AI systems, backend architecture, and applied machine learning. Experienced in full-stack annotation workflows and AI-assisted labeling tools during a 6-month Software Engineer (AI) internship at amasQS.ai. Published at IEEE ICAAIC 2025 and shipped production-style projects in finance automation, real-time task management, and computer vision using Next.js, Node.js, FastAPI, PostgreSQL, MongoDB, Supabase, OpenCV, and TensorFlow.

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, TypeScript, C++
- Frontend:** React, Next.js, Redux, Tailwind CSS
- Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT authentication, WebSockets
- Databases:** PostgreSQL (SQL), MongoDB, Supabase
- AI/ML:** TensorFlow, OpenCV, MediaPipe, Scikit-learn, Gemini AI, LLM applications
- DevOps & Cloud:** Docker, GitHub Actions, CI/CD, AWS, Vercel
- Tools:** Git, Postman, Figma, Kali Linux

EXPERIENCE

- Software Engineer Intern (AI) – amasQIS.ai** Mar 2025 – Sept 2025
- Engineered 5 annotation interfaces for image labeling and text classification, supporting 1,000+ annotator sessions per day
 - Improved dataset quality by adding inter-annotator agreement checks and confidence-score filtering, which raised internal quality scores by 30%
 - Integrated AI-assisted pre-labeling into 2 interfaces using Hugging Face model outputs as default label suggestions, reducing annotator time per item by 40% and cutting dataset preparation time across labeling pipelines
- Web Developer Intern – Zidio Task Management** Feb 2025 – Mar 2025
- Delivered a MERN task management platform in a 1-month sprint and supported 50+ concurrent users at launch
 - Optimized backend performance by developing 10 REST API endpoints for task and user management, bringing response latency down to 560 ms through query indexing and caching
 - Streamlined task views with react-window for virtualized list rendering, eliminating visible lag on lists longer than 20 items

PROJECTS

- FINOVA - AI-Powered Finance Tracker | finovaai-eight.vercel.app** Next.js, Supabase, Gemini AI, Arcjet
- Designed the transaction data model in Supabase with row-level security policies, ensuring each user can only read, write, and delete their own financial records with zero cross-user data exposure
 - Developed receipt scanning with OpenCV preprocessing and Gemini Vision extraction, reducing manual entry by about 75%
 - Configured Clerk authentication, Arcjet rate limiting on financial endpoints, and Resend alerts for spending notifications
- WorkQuanta - Productivity Task Manager | github.com/kalicharm21/Work-Quanta** React, Redux, MongoDB, WebSockets
- Implemented a priority-based task scheduler so earlier-deadline and higher-priority work is processed first
 - Replaced polling with WebSocket push events to reduce wasted requests and make updates feel more real-time
 - Architected REST APIs with JWT auth, role-based access control, and per-user filtering; added rule-based auto-scheduling, reducing manual processing by 40%
- Virtual Mouse - Gesture Control System | github.com/kalicharm21/gesturemouse** OpenCV, MediaPipe, Python, Node.js
- Tracked 21 hand landmarks in real time with MediaPipe Hands at 30 fps and mapped cursor movement to screen coordinates using a screen-calibrated linear transform, enabling smooth and responsive cursor control without any physical device
 - Tuned fingertip-distance thresholds to keep false positives below 5% in standard indoor lighting
 - Reached over 90% gesture recognition accuracy with sub-50 ms end-to-end latency across 3 operating systems

PUBLICATION

Ishaan Mittal et al. "Clinical Decision Support through Machine Learning: Benchmarking LightGBM and Deep Learning Models for Diabetes Onset Prediction." IEEE ICAAIC 2025. Led model benchmarking with 5-fold cross-validation across LightGBM and LSTM; LightGBM outperformed on F1 (0.79 vs 0.74) with 3x faster inference compared to LSTM and feedforward networks, supporting practical deployment in resource-constrained clinical environments

EDUCATION

- Vellore Institute of Technology** Bhopal, India
- B.Tech in Computer Science and Engineering (AIML) | CGPA: 8.55 Sept 2023 – Oct 2027
- Relevant Coursework: Machine Learning, Deep Learning, Data Structures, Algorithms, Computer Vision

CERTIFICATES & ACTIVITIES

- Graph Theory Programming Camp, AlgoUniversity – top 15% of participants (May 2025)
- Cybersecurity Analyst Certificate, IBM (Jun 2025)
- National Space Hackathon 2025, IIT Delhi (Apr 2025) – competed in systems/AI track
- Vice President, Music & Band Club – ran logistics and production for 15+ events
- Outreach Lead, iOS Club – managed external partnerships, coordinated speaker sessions